Does Self-Play on Text Games Teach Reasoning at Small Scale?

Replicating SPIRAL at 4B and Testing Its Limits at 1.7B

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CMPE 260 — Reinforcement Learning · Project Proposal

Project idea. SPIRAL [1] reports that self-play on zero-sum text games raises Qwen3-4B's math accuracy by 8.6% without curated data or task-specific rewards. We test whether this transfer persists at 1.7B, where a model may memorize a shallow game policy; an identical pipeline at 4B replicates SPIRAL as a positive control. This decides its viability at academic-compute scale.

Data set. No labeled data is needed: the reward is Kuhn Poker win/loss in TextArena [2]. We train Qwen3-1.7B-Base and Qwen3-4B-Base and measure transfer on GSM8K (1,319 problems) and MATH-500 (500).

Steps. We adapt SPIRAL using the UnstableBaselines self-play library: LoRA replaces full-parameter updates [3] on one GPU, and a scripted opponent replaces LLM opponents. Weeks 1–2: estimate each model's minimum detectable effect and check our evaluation against the released Spiral-Qwen3-4B checkpoint. Weeks 2–4: scripted-opponent training, replicating SPIRAL's fixed-opponent ablation and validating the pipeline. Weeks 4–8: twelve REINFORCE self-play runs: 2 scales × 2 advantage estimators (role-conditioned vs. global baseline) × 3 seeds. Weeks 8–10: analysis, report, code and adapter release. If self-play is unstable by week 6, we keep only the scale comparison.

Expected results. We plot win rate and Elo against fixed opponents, plus benchmark accuracy and reasoning-trace length across checkpoints, testing before/after accuracy with a paired bootstrap over problems (95% CIs across seeds). We expect 4B to replicate SPIRAL and 1.7B to win games with weaker transfer and shorter traces; transcript audits separate reasoning from memorization.

References. [1] B. Liu et al., "SPIRAL: Self-Play on Zero-Sum Games Incentivizes Reasoning via Multi-Agent Multi-Turn Reinforcement Learning," arXiv:2506.24119, 2025. [2] L. Guertler et al., "TextArena," arXiv:2504.11442, 2025. [3] S. Wang et al., "Tina: Tiny Reasoning Models via LoRA," arXiv:2504.15777, 2025.

Tentative roles. Anurag Bodapally: training (UnstableBaselines, LoRA, HPC jobs). Shubham Baid: evaluation (variance study, benchmark and Elo pipelines, figures). Syed Hassan Muzammil: baselines and analysis (scripted-opponent runs, statistics, transcript audit).

HPC access. We request COE HPC access: twelve multi-turn self-play runs exceed Colab limits. Each targets SPIRAL's 400 steps; we report per-run GPU-hours after week 2.